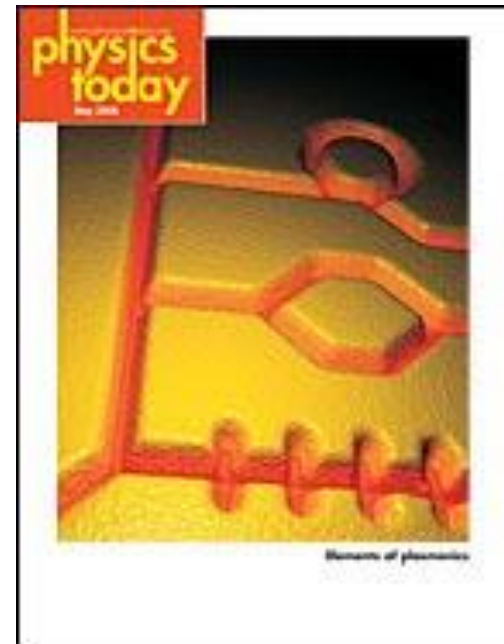


Plasmonic materials and devices

- Circuit model
- Drude model (revision)
- Localized Surface Plasmons
- **Surface Plasmon Polaritons**



Plasmons

Plasma = gas of free charge carriers, e.g. the carriers in a conductor

‘on’ = wavicle

A surface plasmon is a

- collective oscillation of charge carriers near a surface
- a quasiparticle hybrid between electron and photon
- an evanescent wave on a conducting surface...

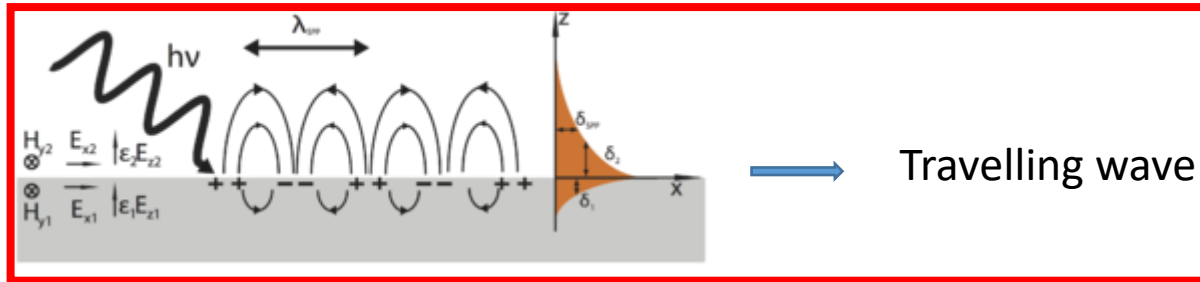
There are two limiting cases of plasmons

- ❖ **LSP** = **localized surface plasmon** (e.g. on a small particle)
- ❖ **SPP** = **surface plasmon polariton** (e.g. on a metal film)

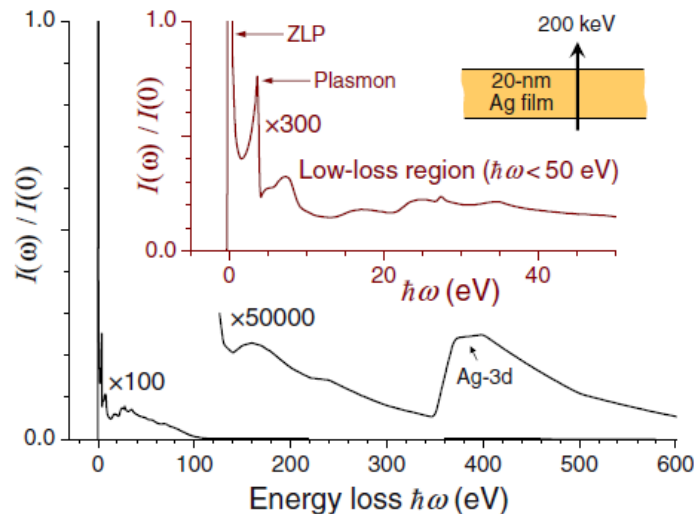
The distinction can become less obvious in-between these two cases

Surface plasmon polariton (SPP)

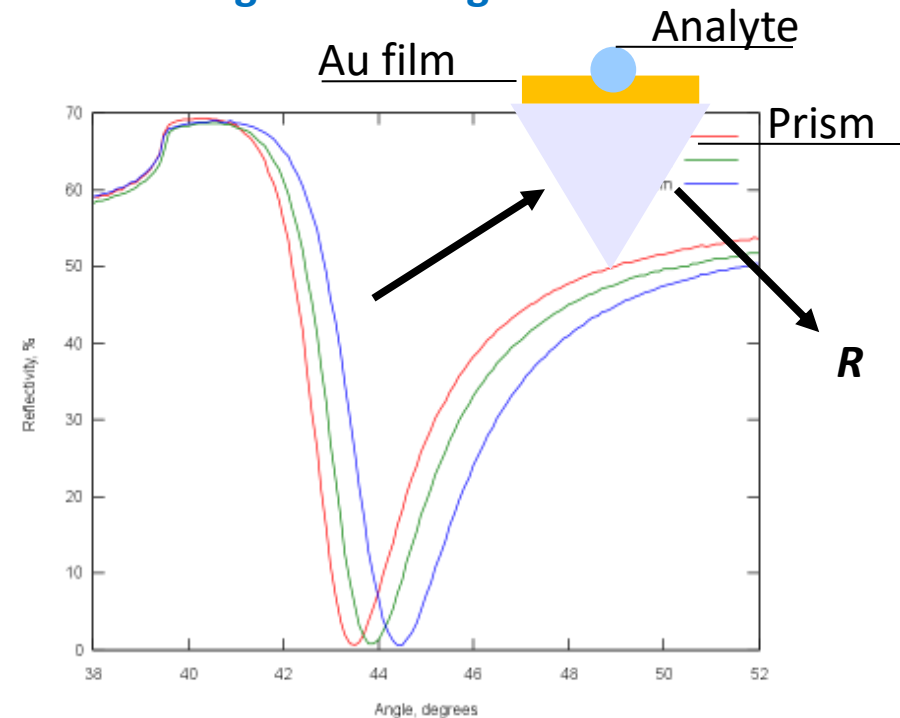
Predicted and observed around 1960s



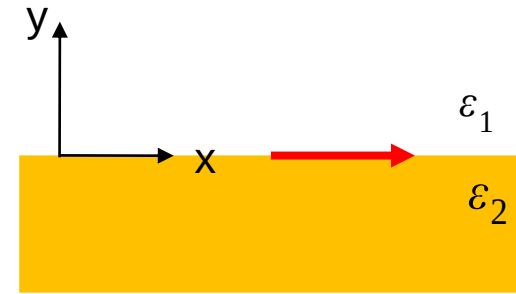
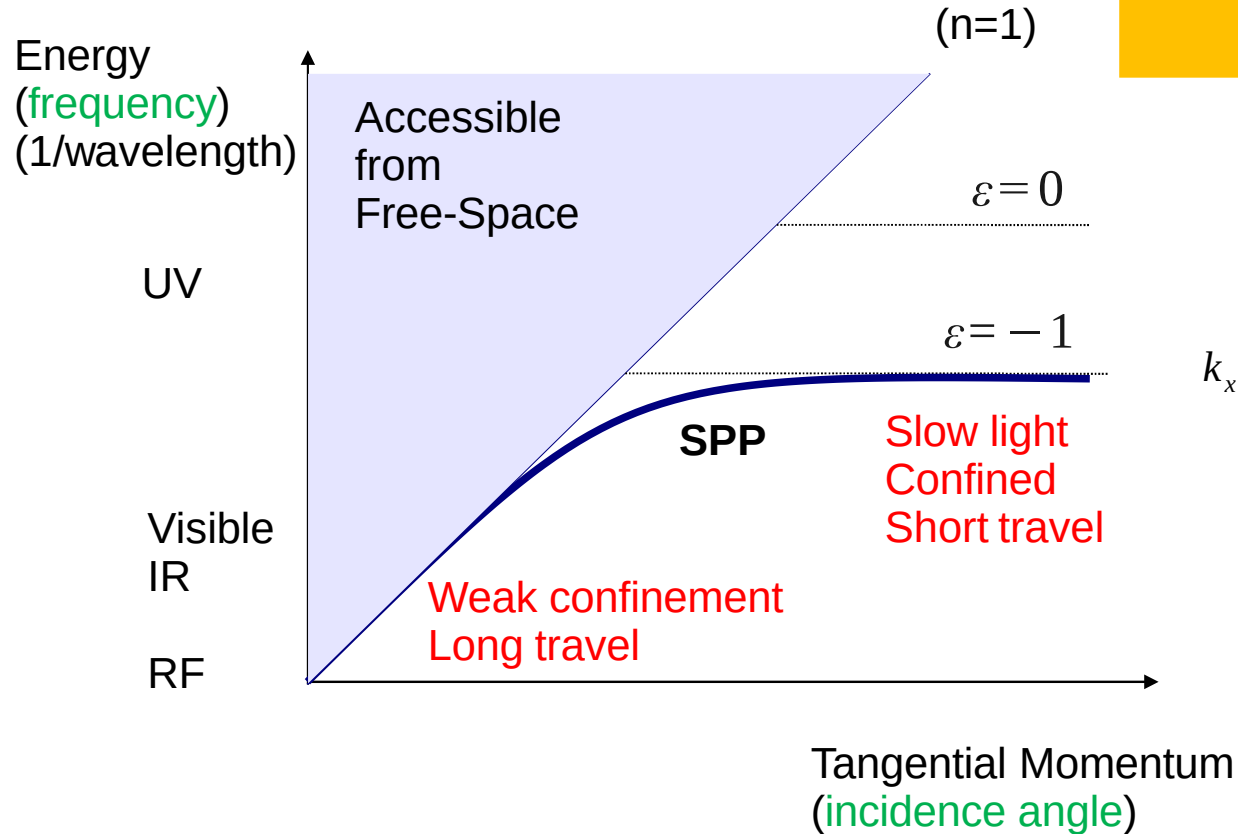
Electron transmission loss



Sensing: R "missing" from TIR



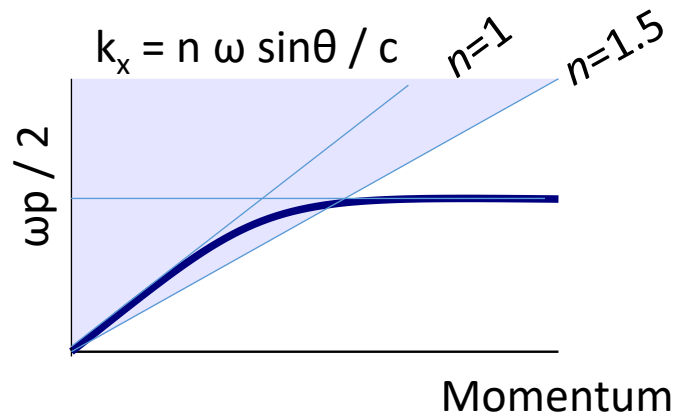
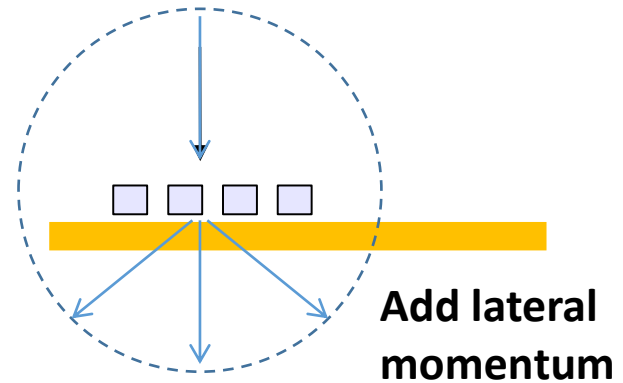
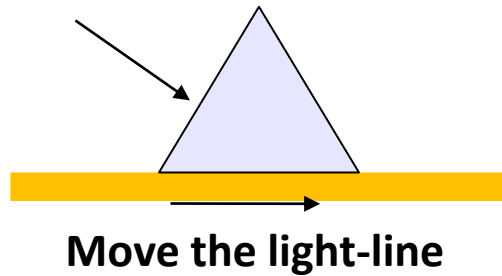
Dispersion - SPP



SPP doesn't intersect light-line = no coupling

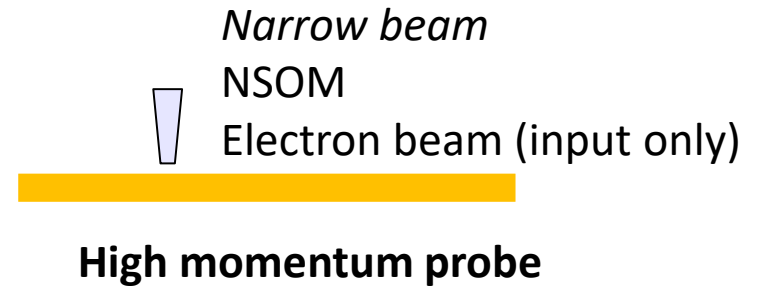
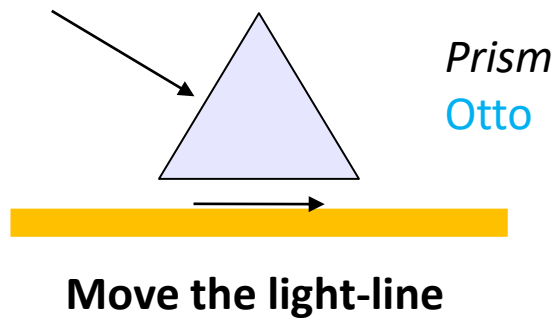
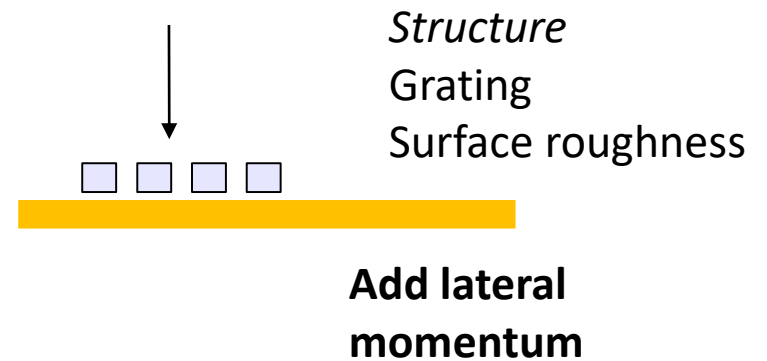
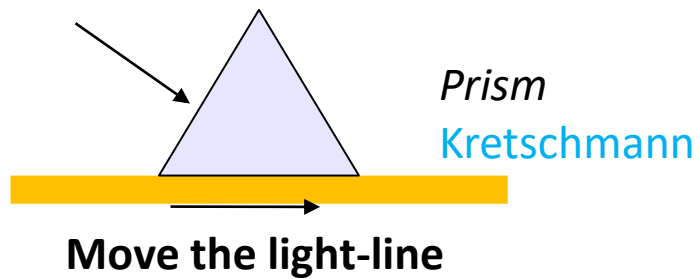
Coupling

Need to address momentum mismatch



If $n=1.5$, what's the *minimum* internal angle to couple to SPP?

Coupling



(Short) derivation of mode

Two media

Planes waves “p” polarization

“No” incident wave = trapped mode very large

Projection of
orthogonal k & E

$$E_{1x} = \frac{k_{1y}}{k_1} E_1 = \sqrt{1 - \left(\frac{k_x}{k_1} \right)^2} E_1$$

Transverse EM

$$E_{1x} = \sqrt{1 - \left(\frac{k_x}{\sqrt{\epsilon_1 \mu_1} \omega} \right)^2} \sqrt{\frac{\mu_1}{\epsilon_1}} H_1$$

Maxwell

Boundary cond.

$$E_{1x} = E_{2x} \quad H_1 = H_2$$

Substitute
& non-mag.

$$\sqrt{1 - \left(\frac{k_x}{\sqrt{\epsilon_1 \mu_0} \omega} \right)^2} \sqrt{\frac{\mu_0}{\epsilon_1}} = \sqrt{1 - \left(\frac{k_x}{\sqrt{\epsilon_2 \mu_0} \omega} \right)^2} \sqrt{\frac{\mu_0}{\epsilon_2}}$$

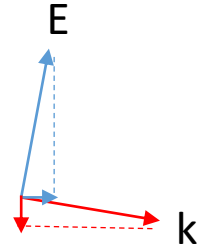
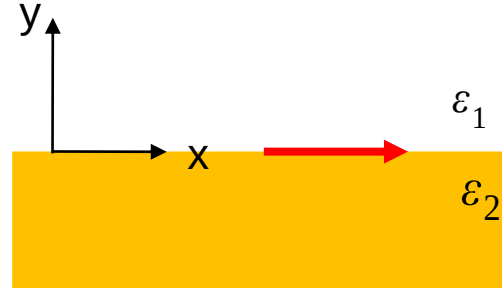
Square,
rearrange

$$(\mu_0 \omega^2 \epsilon_1 - k_x^2) \epsilon_2^2 = (\mu_0 \omega^2 \epsilon_2 - k_x^2) \epsilon_1^2$$

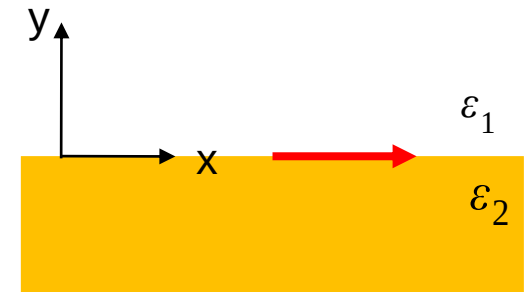
Solve for k

$$k_x = \omega \sqrt{\mu_0 \frac{\epsilon_1 \epsilon_2^2 - \epsilon_2 \epsilon_1^2}{\epsilon_2^2 - \epsilon_1^2}} = \omega \sqrt{\mu_0 \frac{\epsilon_1 \epsilon_2}{\epsilon_2 + \epsilon_1}} = \frac{\omega}{c} \sqrt{\frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2}{\tilde{\epsilon}_2 + \tilde{\epsilon}_1}}$$

Relative
permittivities



Interpretation



$$k_x = \frac{\omega}{c} \sqrt{\frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2}{\tilde{\epsilon}_2 + \tilde{\epsilon}_1}}$$

Tangential k: want real (propagating)

$$k_{1y} = \sqrt{k_1^2 - k_x^2} = \frac{\omega}{c} \sqrt{\tilde{\epsilon}_1 - \frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2}{\tilde{\epsilon}_1 + \tilde{\epsilon}_2}} = \frac{\omega}{c} \frac{\tilde{\epsilon}_1}{\sqrt{\tilde{\epsilon}_1 + \tilde{\epsilon}_2}} \quad \text{Normal: want imag (trap)}$$

- * Only **opposite sign permittivities** give evanescent waves
- * This is provided by metal:dielectric combination
- * **Normal** part gives us confinement (**large imaginary k means tight confinement**)
- * Tangential part gives propagation length (must allow complex permittivities)...

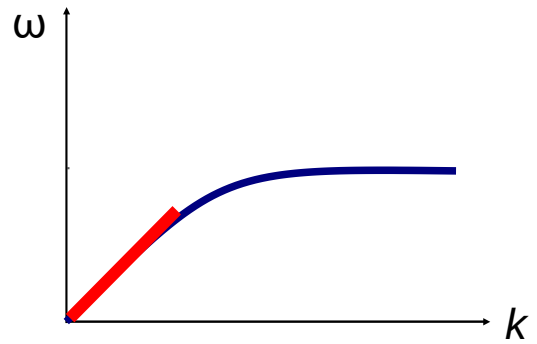
$$\epsilon_2 = \epsilon_2' + i\epsilon_2''$$

$$k_x = \frac{\omega}{c} \sqrt{\frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2}{\tilde{\epsilon}_1 + \tilde{\epsilon}_2}} = \frac{\omega}{c} \sqrt{\frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2'}{\tilde{\epsilon}_1 + \tilde{\epsilon}_2'}} \left[1 + i \frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2''}{2(\tilde{\epsilon}_1 + \tilde{\epsilon}_2') \tilde{\epsilon}_2'} \right]$$

Assuming:
Small imaginary permittivity

Small imaginary wave-vector
Long propagation length
Light like

Drude: light-like



$$k_x = \frac{\omega}{c} \sqrt{\frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2}{\tilde{\epsilon}_1 + \tilde{\epsilon}_2}} = \frac{\omega}{c} \sqrt{\frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2'}{\tilde{\epsilon}_1 + \tilde{\epsilon}_2'}} \left[1 + i \frac{\tilde{\epsilon}_1 \tilde{\epsilon}_2''}{2(\tilde{\epsilon}_1 + \tilde{\epsilon}_2') \tilde{\epsilon}_2'} \right]$$

$$\tilde{\epsilon}_2 \approx -\frac{\omega_p^2}{\omega^2} + i \frac{\omega_p^2 \gamma}{\omega^3} \quad \gamma \ll \omega \ll \omega_p \quad \tilde{\epsilon}_1 \rightarrow 1$$

Light-like

$$k_x \sim \frac{\omega}{c} \left[1 + i \frac{\gamma \omega}{2 \omega_p^2} \right]$$

The imaginary part is not very accurate unless
 $\omega \ll \omega_p$

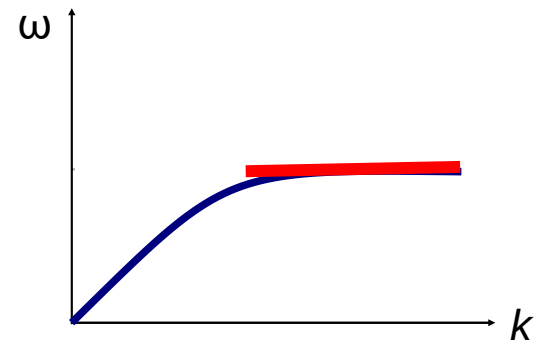
Drude: slow light

$$\omega \sim \omega_p / \sqrt{2} \quad \text{Slow light}$$

$$\tilde{\epsilon}_2 \approx -1 + i \frac{2\sqrt{2}\gamma}{\omega_p}$$

We can't use the previous expression for k (here is too lossy)...
do series expansion, then *very* approximately

$$k_x \sim \frac{\omega_p}{c 2^{7/4} \sqrt{\gamma / \omega_p}} [1 + i]$$

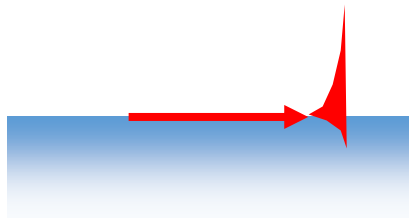


$$k_x \sim \frac{\omega}{c} \left[1 + i \frac{\gamma \omega}{2 \omega_p^2} \right] \quad k_x \sim \frac{\omega_p}{c 2^{7/4} \sqrt{\gamma / \omega_p}} [1 + i]$$

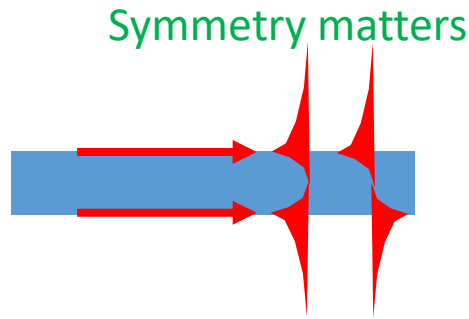
Estimate the propagation length $1/\text{Im}(k)$ for Al @ 1.55 μm and at the “slow light” limit

Thin films stacks

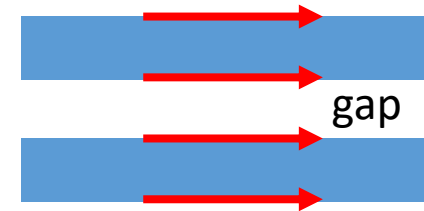
Plasmons exist at each surface: they hybridize



Single frequency



Film: two frequencies

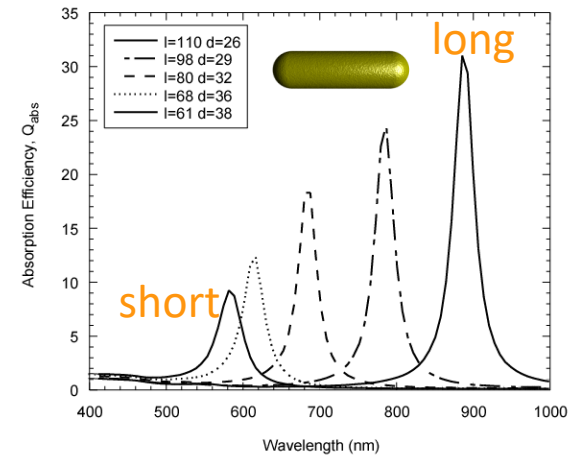
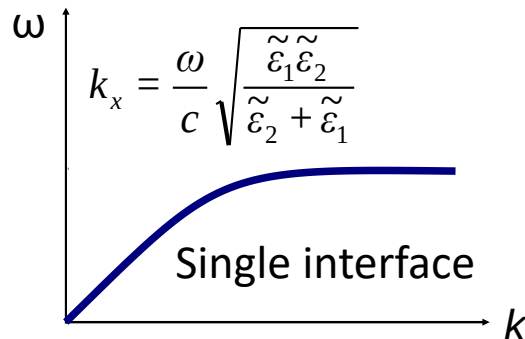
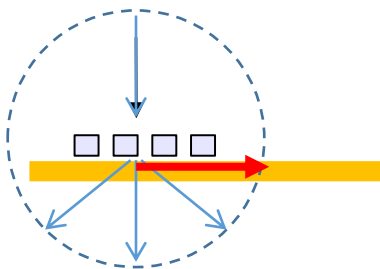


Four frequencies

The concentration of field in the metal affects the loss
If we can trap the plasmon in a **gap** it should have lower loss

Summary

- Plasmons = surface resonance of conductor
- Drude model useful for basic material response
- LSP: small particle traps
 - Shape most useful for tuning
 - Electron forces key to understanding
 - Modelled by quasistatic resonance
- SPP: thin film waveguides
 - Needs special coupling (prism/gratings)
 - Highly dispersive: light-like → slow light
 - Thickness determines properties



$$\alpha = V \sum_m \frac{A_m}{S_m + 1/\chi}$$

